AI Based Crop Recommendation System for Farmers Using ML

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Abstract

Agriculture plays a vital role in economic growth; however, farmers often face challenges in selecting suitable crops due to changing climatic conditions, soil variability, and the lack of data-driven guidance. This project presents an AIbased Crop Recommendation System that assists farmers in choosing appropriate crops based on soil type, land area (square feet), climatic factors, and crop characteristics. The proposed system collects historical agricultural data such as soil nutrients, moisture levels, temperature, rainfall, and past crop yields. A neural network-based regression algorithm is employed to model the relationship between environmental parameters and crop performance. The trained regression model predicts crop suitability scores, based on which optimal crops are recommended for specific land and soil conditions. The system enables informed decision-making, improves agricultural productivity, reduces the risk of crop failure, and ensures efficient utilization of resources. This AI-driven approach supports sustainable farming practices and enhances intelligent decision support in modern agriculture.

1.Introduction

Agriculture is the backbone of many economies, especially in developing countries, yet farmers often struggle to choose the right crop for their land. Factors such as unpredictable climate conditions, soil fertility variations, and limited access to expert guidance make crop selection a challenging task. Traditional farming practices rely heavily on experience and intuition, which may not always yield optimal results. With the rapid advancement of Artificial Intelligence (AI), data-driven approaches can now support farmers in making better crop decisions.

Crop selection depends on multiple parameters, including soil type, land area, temperature, rainfall, and nutrient availability. Analyzing these factors manually is time-consuming and prone to errors. AI-based systems can process large volumes of historical and real-time agricultural data efficiently. By identifying hidden patterns and correlations, AI helps overcome uncertainties caused by climate change and environmental variability.

The AI-based Crop Recommendation System collects data such as soil nutrients (nitrogen, phosphorus, potassium), moisture levels, climatic conditions, and land size measured in square feet. This data is preprocessed and fed into machine learning models like regression, decision trees, or neural networks. The trained model evaluates the suitability of different crops and generates personalized recommendations for farmers based on their specific land and soil conditions.

Machine learning plays a crucial role in improving the accuracy of crop recommendations. Models are trained using historical crop yield data to understand how environmental factors influence crop growth. Neural networks, in particular, can capture complex non-linear relationships between inputs and outputs. As more data is collected over time, the system continuously learns and improves its prediction performance.

This intelligent system enables farmers to make informed decisions before investing time and resources into cultivation. By selecting crops best suited to their land, farmers can increase productivity, reduce the risk of crop failure, and minimize unnecessary use of water, fertilizers, and pesticides. The system is especially beneficial for small-scale farmers who lack access to agricultural experts.

AI-driven crop recommendation promotes sustainable farming practices by optimizing resource utilization and preventing soil degradation. Growing suitable crops helps maintain soil health and reduces environmental impact. Economically, higher yields and reduced losses lead to increased income stability for farmers, contributing to rural development and food security.

In conclusion, the AI-Based Crop Recommendation System offers a smart and practical solution to modern agricultural challenges. By combining soil, climate, and land data with advanced machine learning techniques, the system supports accurate and reliable crop selection. This approach not only enhances agricultural productivity but also encourages sustainable and technology-driven farming, paving the way for a smarter future in agriculture

2.Literature Review

1. Sharma et al. (2018)

Sharma et al. presented a machine learning–based crop recommendation system that analyzed soil nutrients, rainfall, and temperature data. The study emphasized the role of historical agricultural datasets in predicting suitable crops. Various preprocessing techniques were applied to remove noise and improve data quality. Classification algorithms were used to map environmental conditions to crop types. The system demonstrated improved decision-making for farmers compared to traditional methods. However, limited dataset size affected the overall prediction accuracy.

2. Patel et al. (2019)

Patel et al. developed a decision-support system for crop selection using algorithms such as Naïve Bayes and Decision Trees. The model considered soil fertility parameters like nitrogen, phosphorus, and potassium levels. Rainfall data was also included to enhance crop suitability analysis. The results showed acceptable accuracy for crop recommendation in controlled conditions. The study highlighted the importance of data-driven agriculture.

However, real-time weather changes were not incorporated into the system.

3. Kumar et al. (2020)

Kumar et al. proposed a crop recommendation framework using regression and ensemble learning techniques. The system analyzed historical yield data along with climatic parameters to predict crop performance. Multiple models were trained and compared to improve reliability. The ensemble approach increased accuracy over single-model methods. The study demonstrated the usefulness of machine learning in agricultural planning. Nevertheless, the model required large datasets and high computational resources.

4. Reddy et al. (2020)

Reddy et al. focused on combining IoT-based sensor data with machine learning algorithms for crop recommendation. Sensors were used to collect real-time soil moisture and temperature values. The collected data was transmitted to a cloud platform for processing. Machine learning models analyzed the data to suggest suitable crops. The system improved the accuracy of recommendations through real-time monitoring. However, increased hardware cost was a major limitation.

5. Singh et al. (2021)

Singh et al. introduced a neural network–based crop recommendation system considering soil type, land area, and climatic conditions. The neural network model effectively captured complex relationships between input parameters and crop yield. Experimental results showed higher accuracy compared to traditional classifiers. The system supported better crop planning and reduced crop failure risk. It was particularly useful for precision agriculture applications. The drawback was increased training time and computational complexity.

6. Verma et al. (2022)

Verma et al. proposed a cloud-based AI crop advisory system to provide scalable recommendations to farmers. The system stored agricultural data on cloud servers for easy access and processing. Farmers could receive crop suggestions using mobile or web applications. Machine learning models were used to analyze soil and weather data. The platform improved accessibility and usability. However, poor internet connectivity in rural areas affected system performance.

7. Das et al. (2023)

Das et al. explored deep learning techniques for intelligent crop recommendation under changing climate conditions. The model adapted to seasonal and environmental variations using continuous learning. Large-scale datasets were used to train deep neural networks. The system achieved high accuracy in crop suitability prediction. The study highlighted the role of AI in climate-resilient agriculture. However, frequent data updates were required to maintain accuracy.

3.Research Gap

From the reviewed literature, it is observed that most existing AI-based crop recommendation systems rely on limited datasets, static environmental parameters, or isolated factors such as soil nutrients or rainfall. Many studies do not integrate real-time data, land area constraints (in square feet), and dynamic climate variations into a single unified model. Additionally, the majority of systems focus on improving prediction accuracy but give less attention to practical usability for small-scale farmers, including cost, accessibility, and adaptability to local conditions. Hence, there is a clear research gap in developing an integrated, real-time, and farmer-friendly AI crop recommendation system that combines soil properties, land size, and climatic factors while ensuring scalability and sustainability.

4.Proposed system

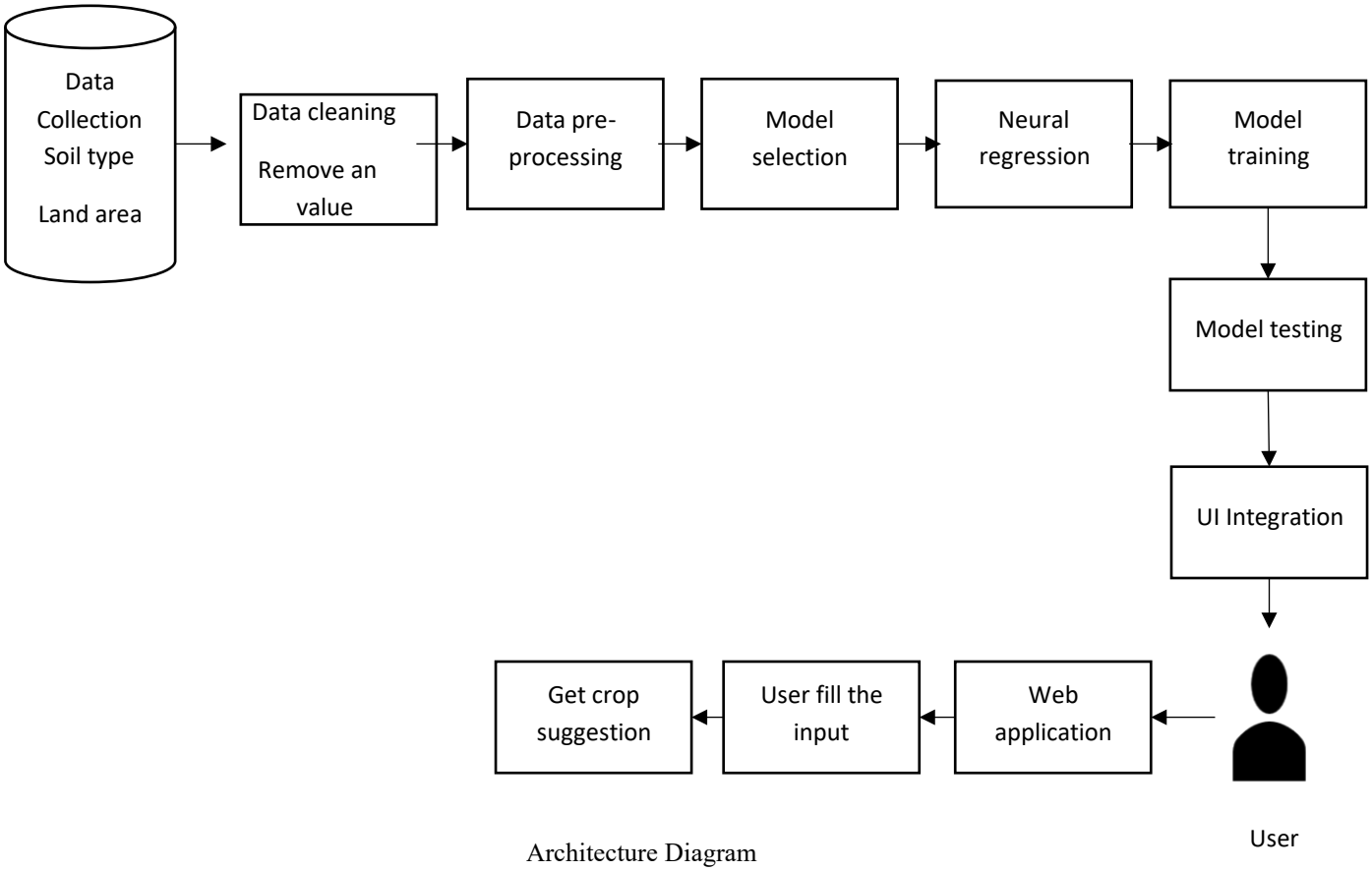
The proposed AI-Based Crop Recommendation System is designed to provide accurate and personalized crop suggestions for farmers by integrating soil characteristics, land area in square feet, and climatic conditions. The system aims to overcome the limitations of traditional farming methods by using data-driven decision-making. It focuses on simplicity and accessibility so that even smallscale farmers can effectively use the system. By combining multiple parameters, the system ensures more reliable and practical crop recommendations.

In the proposed system, agricultural data such as soil nutrients, moisture levels, temperature, rainfall, and land size is collected either from farmers or through sensors and agricultural databases. This data is preprocessed to handle missing values and normalize inputs. Machine learning algorithms, including regression models and neural networks, are trained using historical crop yield data. The trained model learns complex relationships between environmental factors and crop suitability.

Once trained, the system analyzes user-provided inputs and predicts the most suitable crops for the given land conditions. The recommendation output includes a list of crops ranked based on suitability and expected performance. The system can also be updated periodically with new data to improve accuracy over time. This adaptive learning capability helps the model respond to changing climate patterns and soil conditions.

Overall, the proposed system enhances agricultural productivity by reducing crop failure risks and optimizing resource utilization. It supports sustainable farming practices by encouraging the selection of crops best suited to local conditions. The system is designed to be scalable and can be deployed as a web or mobile application for wider reach. Thus, the proposed AI-based solution serves as a practical and efficient tool for modern precision agriculture.

Architecture Diagram



Architecture Diagram

Working Principle

The AI-Based Crop Recommendation System works by collecting key

agricultural parameters such as soil type, nutrient levels, land area in square feet, temperature, and rainfall. This input data is first preprocessed to remove noise and handle missing values. The processed data is then passed to trained machine learning models, such as regression or neural networks, which analyze the relationships between environmental conditions and crop suitability. Based on the model's prediction, the system recommends the most suitable crops for the given land conditions. The output helps farmers make informed crop selection decisions before cultivation.

Advantages

- Provides accurate and data-driven crop recommendations.
- Reduces crop failure risk caused by unsuitable crop selection.
- Considers multiple parameters such as soil type, land area, and climate.
- Improves crop yield and overall farm productivity.
- Optimizes use of water, fertilizers, and other resources.
- Saves time and effort compared to traditional trial-and-error methods.
- Supports sustainable and eco-friendly farming practices.
- Beneficial for small and marginal farmers with limited expert access.
- Helps farmers make informed decisions before cultivation.
- Can be easily deployed as a web or mobile application.

Methodology

To validate and evaluate the proposed system, agricultural data sets from the past, including details of the soil, climate, and crop yields, were used. An 80:20 data division for training and testing purposes was used.

Input Data

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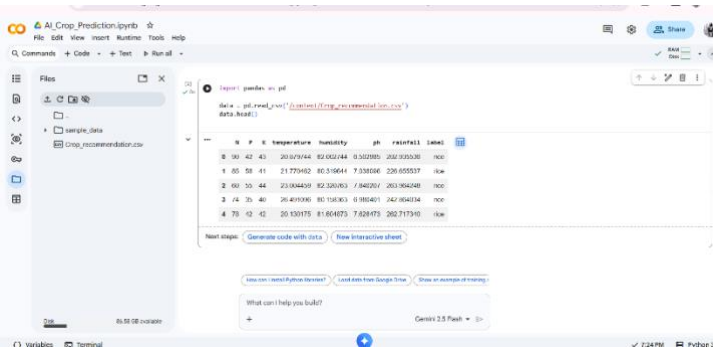
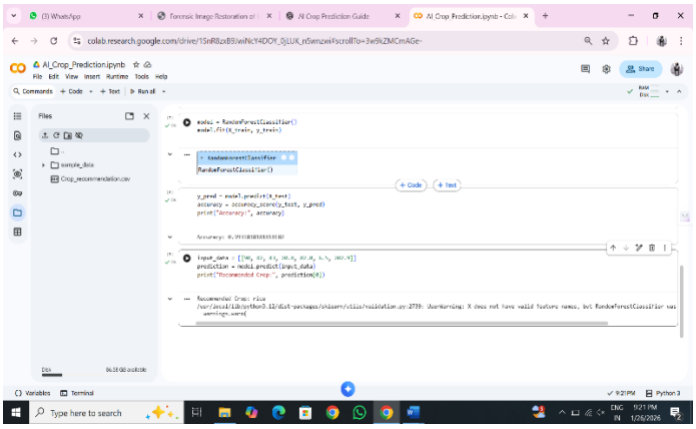


The input data includes nutrient content of soil (N, P, K), soil pH, land area in terms of square feet, temperature, rainfall. These factors are those that generally affect the growth of crops.

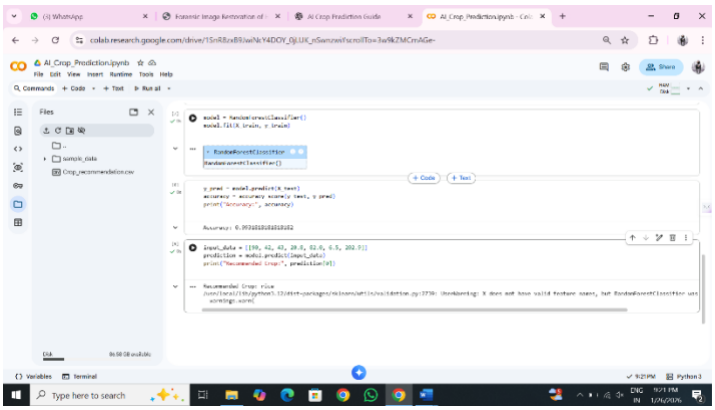
Prepocessing

Normalization was applied to all input features to enhance convergence. Incomplete information was addressed through statistical imputations. Categorical information was encoded numerically.

Model Training And Testing

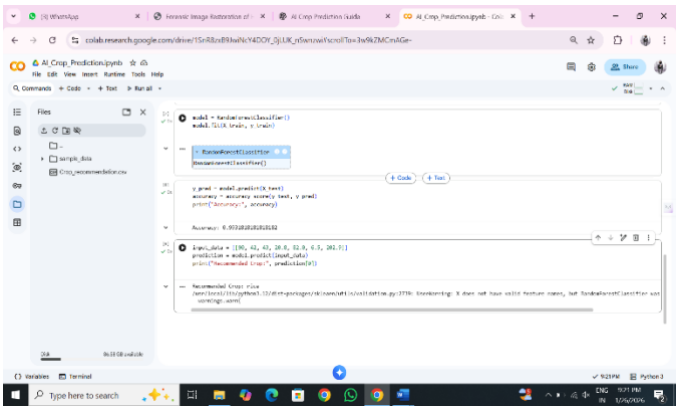


A neural network-based regression model was used for predicting crop suitability score, and the test of the model performance was carried out using accuracy and mean squared error (MSE). The model trained showed good prediction capability under varying input conditions.



Ouput Results

It produces a list of suitable crops that could grow in the provided land. There are improved accuracy results and better adherence to the best agricultural practices.



conclusion

The AI-Based Crop Recommendation System offers an effective solution to modern agricultural challenges by combining artificial intelligence with agricultural data. By analyzing soil, land size, and climatic conditions, the system provides reliable crop suggestions tailored to individual farms. This approach improves decision-making, enhances productivity, and reduces environmental impact. The system demonstrates the potential of AI in transforming traditional agriculture into smart and sustainable farming.

Future work

In future, the system can be enhanced by integrating real-time IoT sensor data for more accurate predictions. Advanced deep learning models can be used to further improve recommendation accuracy. The system can also include fertilizer and irrigation recommendations along with crop suggestions. Mobile app deployment with regional language support will increase usability among farmers. Integration with market price prediction and government advisory services can make the system more comprehensive

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